$$
\begin{aligned}
& C_{dd}{}^{i1\_i2\_i3\_i4\_} \to \frac{1}{16\pi^2}\ \Bigg(\frac{1}{648}\ \frac{1}{m_\Phi^2}\ s_\gamma^2\ \Big(\overline{y_d}^{pi3}\ \big(y_d^{pi4}\ (g_1^2+12\ g_3^2)\ \delta_{i1i2}-36\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(-36\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}+y_d^{pi2}\ \big(-162\ c_\gamma^2\ \overline{y_d}^{ri3}\ y_d^{ri4}+(g_1^2+12\ g_3^2)\ \delta_{i3i4}\big)\big)\Big)+ \\
& \frac{1}{18}\ g_3^4\ (3\ \delta_{i1i4}\ \delta_{i2i3}-\delta_{i1i2}\ \delta_{i3i4})\ LF_{3,0}[m_3]+\frac{1}{12}\ g_3^4\ (3\ \delta_{i1i4}\ \delta_{i2i3}-\delta_{i1i2}\ \delta_{i3i4})\ LF_{4,-1}[m_3]+ \\
& \frac{4}{45}\ g_3^4\ (-3\ \delta_{i1i4}\ \delta_{i2i3}+\delta_{i1i2}\ \delta_{i3i4})\ LF_{5,-2}[m_3]+ \\
& \frac{1}{486}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(4\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,0}[m_{\tilde{d}}^{p}]- \\
& \frac{5}{1296}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(4\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,-1}[m_{\tilde{d}}^{p}]+ \\
& \frac{2}{1215}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(4\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{5,-2}[m_{\tilde{d}}^{p}]+ \\
& \frac{2}{81}\ \sum_p g_1^4\ LF_{3,0}[m_{\tilde{e}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}-\frac{5}{108}\ \sum_p g_1^4\ LF_{4,-1}[m_{\tilde{e}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{8}{405}\ \sum_p g_1^4\ LF_{5,-2}[m_{\tilde{e}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{1}{81}\ \sum_p g_1^4\ LF_{3,0}[m_{\tilde{l}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}- \\
& \frac{5}{216}\ \sum_p g_1^4\ LF_{4,-1}[m_{\tilde{l}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{4}{405}\ \sum_p g_1^4\ LF_{5,-2}[m_{\tilde{l}}^{p}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{243}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,0}[m_{\tilde{q}}^{p}]- \\
& \frac{5}{648}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,-1}[m_{\tilde{q}}^{p}]+ \\
& \frac{4}{1215}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{5,-2}[m_{\tilde{q}}^{p}]+ \\
& \frac{1}{486}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(16\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,0}[m_{\tilde{u}}^{p}]- \\
& \frac{5}{1296}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(16\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,-1}[m_{\tilde{u}}^{p}]+ \\
& \frac{2}{1215}\ \sum_p\ \big(27\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+(16\ g_1^4-9\ g_3^4)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{5,-2}[m_{\tilde{u}}^{p}]+ \\
& \frac{1}{54}\ s_\gamma^2\ \Big(\overline{y_d}^{pi3}\ \big(y_d^{pi4}\ (g_1^2+3\ g_3^2)\ \delta_{i1i2}-9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(-9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}+y_d^{pi2}\ \big(27\ c_\gamma^2\ \overline{y_d}^{ri3}\ y_d^{ri4}+(g_1^2+3\ g_3^2)\ \delta_{i3i4}\big)\big)\Big)\ LF_{1,2}[m_\Phi]+ \\
& \frac{1}{36}\ s_\gamma^2\ \big(-g_1^2\ \overline{y_d}^{pi3}\ y_d^{pi4}\ \delta_{i1i2}+\overline{y_d}^{pi1}\ y_d^{pi2}\ (9\ s_\gamma^2\ \overline{y_d}^{ri3}\ y_d^{ri4}-g_1^2\ \delta_{i3i4})\big)\ LF_{2,1}[m_\Phi]+ \\
& \frac{1}{324}\ g_1^2\ \big(9\ s_\gamma^2\ \overline{y_d}^{pi3}\ y_d^{pi4}\ \delta_{i1i2}+(9\ s_\gamma^2\ \overline{y_d}^{pi1}\ y_d^{pi2}+4\ g_1^2\ \delta_{i1i2})\ \delta_{i3i4}\big)\ LF_{3,0}[m_\Phi]- \\
& \frac{5}{216}\ g_1^4\ LF_{4,-1}[m_\Phi]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{4}{405}\ g_1^4\ LF_{5,-2}[m_\Phi]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{81}\ g_1^4\ LF_{3,0}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{1}{54}\ g_1^4\ LF_{4,-1}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4}-\frac{8}{405}\ g_1^4\ LF_{5,-2}[\tilde{\mu}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_1,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_1,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_1,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_1,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_1,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_1,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_1,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_1,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_1,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_1,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_1,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_1,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_1,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_1,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_1,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_1,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_3,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_3,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1296}\ \big(-225\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (-32\ g_1^2+75\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_3,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{162}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_3,\ m_{\tilde{d}}^{i1}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_3,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_3,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1296}\ \big(-225\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (-32\ g_1^2+75\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_3,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{162}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_3,\ m_{\tilde{d}}^{i2}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_3,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_3,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1296}\ \big(-225\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (-32\ g_1^2+75\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_3,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{162}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_3,\ m_{\tilde{d}}^{i3}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_3,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,2,-1}[m_3,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{1296}\ \big(-225\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (-32\ g_1^2+75\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_3,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{162}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{4,1,-2}[m_3,\ m_{\tilde{d}}^{i4}]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i1},\ m_1]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i1},\ m_1]+ \\
& \frac{1}{648}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}-g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i1},\ m_3]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i1},\ m_3]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i2},\ m_1]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i2},\ m_1]+ \\
& \frac{1}{648}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}-g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i2},\ m_3]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i2},\ m_3]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i3},\ m_1]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i3},\ m_1]+ \\
& \frac{1}{648}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}-g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i3},\ m_3]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i3},\ m_3]+ \\
& \frac{1}{972}\ g_1^2\ \big(-9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(-2\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i4},\ m_1]+ \\
& \frac{1}{1944}\ g_1^2\ \big(9\ g_3^2\ \delta_{i1i4}\ \delta_{i2i3}+(2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i4},\ m_1]+ \\
& \frac{1}{648}\ \big(9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}-g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{2,1,0}[m_{\tilde{d}}^{i4},\ m_3]+ \\
& \frac{1}{1296}\ \big(-9\ g_3^4\ \delta_{i1i4}\ \delta_{i2i3}+g_3^2\ (16\ g_1^2+3\ g_3^2)\ \delta_{i1i2}\ \delta_{i3i4}\big)\ LF_{3,1,-1}[m_{\tilde{d}}^{i4},\ m_3]+ \\
& \frac{1}{54}\ \Big(\overline{y_d}^{pi3}\ \big(y_d^{pi4}\ (g_1^2+3\ g_3^2)\ \delta_{i1i2}-9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(-9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}+y_d^{pi2}\ (g_1^2+3\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{2,1,0}[m_{\tilde{q}}^{p},\ \tilde{\mu}]+ \\
& \frac{1}{108}\ \Big(\overline{y_d}^{pi3}\ \big(-y_d^{pi4}\ (g_1^2+3\ g_3^2)\ \delta_{i1i2}+9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}-y_d^{pi2}\ (g_1^2+3\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{2,2,-1}[m_{\tilde{q}}^{p},\ \tilde{\mu}]+ \\
& \frac{1}{108}\ \Big(\overline{y_d}^{pi3}\ \big(-y_d^{pi4}\ (g_1^2+3\ g_3^2)\ \delta_{i1i2}+9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}-y_d^{pi2}\ (g_1^2+3\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{3,1,-1}[m_{\tilde{q}}^{p},\ \tilde{\mu}]+ \\
& \frac{1}{108}\ \Big(\overline{y_d}^{pi3}\ \big(-y_d^{pi4}\ (g_1^2+3\ g_3^2)\ \delta_{i1i2}+9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}-y_d^{pi2}\ (g_1^2+3\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{2,1,0}[\tilde{\mu},\ m_{\tilde{q}}^{p}]+ \\
& \frac{1}{108}\ \Big(\overline{y_d}^{pi3}\ \big(y_d^{pi4}\ (-7\ g_1^2+6\ g_3^2)\ \delta_{i1i2}-18\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(-18\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}+y_d^{pi2}\ (-7\ g_1^2+6\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{3,1,-1}[\tilde{\mu},\ m_{\tilde{q}}^{p}]+ \\
& \frac{1}{108}\ \Big(\overline{y_d}^{pi3}\ \big(y_d^{pi4}\ (2\ g_1^2-3\ g_3^2)\ \delta_{i1i2}+9\ g_3^2\ y_d^{pi2}\ \delta_{i1i4}\big)+ \\
&\qquad \overline{y_d}^{pi1}\ \big(9\ g_3^2\ y_d^{pi4}\ \delta_{i2i3}+y_d^{pi2}\ (2\ g_1^2-3\ g_3^2)\ \delta_{i3i4}\big)\Big)\ LF_{4,1,-2}[\tilde{\mu},\ m_{\tilde{q}}^{p}]+ \\
& \frac{1}{324}\ g_1^4\ LF_{2,1,1,-1}[m_1,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{1}{162}\ g_1^4\ m_1^2\ LF_{2,1,1,0}[m_1,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{324}\ g_1^4\ LF_{2,1,1,-1}[m_1,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{1}{162}\ g_1^4\ m_1^2\ LF_{2,1,1,0}[m_1,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{7}{48}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3}-\frac{1}{12}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3}+ \\
& \frac{1}{144}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{5}{36}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{144}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}+\frac{5}{36}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{7}{48}\ g_3^4\ LF_{2,1,1,-1}[m_3,\ m_{\tilde{d}}^{i3},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3}-\frac{1}{12}\ g_3^4\ m_3^2\ LF_{2,1,1,0}[m_3,\ m_{\tilde{d}}^{i3},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3}+ \\
& \frac{1}{4}\ \overline{y_d}^{pi1}\ \overline{y_d}^{ri3}\ y_d^{pi2}\ y_d^{ri4}\ LF_{2,1,1,-1}[\tilde{\mu},\ m_{\tilde{q}}^{p},\ m_{\tilde{q}}^{r}]+ \\
& \frac{1}{36}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3}+ \\
& \frac{1}{18}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i2}]\ \delta_{i1i4}\ \delta_{i2i3}- \\
& \frac{1}{108}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}- \\
& \frac{1}{54}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{d}}^{i1},\ m_{\tilde{d}}^{i3}]\ \delta_{i1i2}\ \delta_{i3i4}- \\
& \frac{1}{108}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}- \\
& \frac{1}{54}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{d}}^{i2},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i2}\ \delta_{i3i4}+ \\
& \frac{1}{36}\ g_1^2\ g_3^2\ LF_{1,1,1,1,-1}[m_1,\ m_3,\ m_{\tilde{d}}^{i3},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3}+ \\
& \frac{1}{18}\ m_1\ m_3\ g_1^2\ g_3^2\ LF_{1,1,1,1,0}[m_1,\ m_3,\ m_{\tilde{d}}^{i3},\ m_{\tilde{d}}^{i4}]\ \delta_{i1i4}\ \delta_{i2i3}\Bigg)
\end{aligned}
$$